



Frequency Reuse with Clustering

◆ Definition

Frequency reuse is a technique in cellular networks where the **same frequency channels are reused in different cells** to improve spectrum efficiency and increase system capacity.

◆ Basic Concept

- The total coverage area is divided into **small cells**
 - Each cell is assigned a set of **frequencies**
 - The same frequencies are reused in **non-adjacent cells** to avoid interference
-

◆ Clustering Concept

- A group of adjacent cells using **different frequency sets** is called a **cluster**
 - Each cluster uses the **complete set of available frequencies**
 - Clusters are **repeated over the entire network**
-

◆ Cluster Size (N)

Cluster size determines how frequencies are distributed.

$$N = i^2 + ij + j^2$$

Where:

- **i, j** = integers
- **N** = number of cells in a cluster



Examples:

- $(i=1, j=1) \rightarrow N = 3$
 - $(i=2, j=1) \rightarrow N = 7$
 - $(i=2, j=2) \rightarrow N = 12$
-

◆ Frequency Reuse Pattern

- Cells using the same frequency are called **co-channel cells**
 - They are separated by a distance to reduce interference
-

◆ Reuse Distance (D)

$$D = \sqrt{3N} \cdot R$$

Where:

- **D** = distance between co-channel cells
 - **R** = cell radius
 - **N** = cluster size
-

◆ Advantages

- Efficient use of limited frequency spectrum
 - Increased network capacity
 - Supports more users
-

◆ Disadvantages

- Co-channel interference
 - Complex frequency planning required
-

Quick Summary (Exam Tip)

- **Cluster** = group of cells with unique frequencies
 - **Frequency reuse** = reuse same frequencies in far cells
 - Larger **N** → **less interference, lower capacity**
 - Smaller **N** → **more capacity, more interference**
-

